

MetaInspector Desktop — Sprint 1 & 2 Progress Report

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Summary

Sprints 1 and 2 are both complete: 10 of 23 project tasks (about 43%) are done, and all 58 automated tests pass.

Sprint	Scope	Branch	Status
1	Core scraping engine (TSK-101 to TSK-105)	feature/sprint1-core-engine	Merged into dev (PR #1)
2	Desktop GUI (TSK-201 to TSK-205)	feature/sprint2-desktop-gui	Pushed, PR to dev awaiting review
3	Google Sheets export + SQLite history	—	Not started
4	100-account speed and edge-case testing	—	Not started
5	.exe packaging and client handover	—	Not started

The main open risk: country and joined-date extraction has not yet been tested with a logged-in checker account.

Sprint 1: Core engine (merged)

The scraping engine under `src/core/` was delivered, reviewed and merged into `dev` as PR #1.

Task	File	What it does
TSK-101	<code>resource_blocker.py</code>	Aborts image, video and font requests per page; keeps HTML, JS, CSS and XHR/GraphQL. Blocked 20–27 assets per live Instagram page load.
TSK-102	<code>extractors.py</code>	Instagram: classifies active / private / not found / session blocked, opens "About this account", polls up to 8 s for joined date and country (English + Turkish).
TSK-103	<code>extractors.py</code>	Threads: finds the menu by SVG position, opens "About this profile", splits joined date and badge (e.g. 100M+).
TSK-104	<code>ban_engine.py</code>	<code>BanLinkEngine</code> : a ban on either platform marks both BANNED.
TSK-105	<code>scraper.py</code>	<code>MultiWorkerScraper</code> : 5 pages, balanced chunks (100 accounts = 5×20), live progress callback, human-like delays, auto-stop if Meta challenges the checker session. CLI with <code>--login</code> .

Sprint 2: Desktop GUI (awaiting review)

The CustomTkinter desktop app under `src/gui/` is pushed on `feature/sprint2-desktop-gui`; launch it with `python main.py`.

Task	File	What it does
TSK-201	<code>app.py</code> , <code>theme.py</code>	1150×750 Cyber Dark window (min 950×600) with header, left panel, live table and status bar; colour and font tokens from <code>design.md</code> .
TSK-202	<code>components/left_panel.py</code>	Username box with live count, <code>.txt</code> import, mode radios (Combined / Instagram Only / Threads Only), Start / Pause / Stop.
TSK-203	<code>components/data_table.py</code>	Results table: ACTIVE green, BANNED red, PRIVATE and BLOCKED amber, NOT_FOUND grey; per-status counts.

Task	File	What it does
TSK-204	<code>bridge.py</code>	Scraper runs on a background thread; callbacks only enqueue events, drained every 100 ms by <code>root.after()</code> , so the window never freezes.
TSK-205	<code>app.py</code> , <code>core/session.py</code>	"Setup Checker Account" opens the login browser on a background thread; header shows Connected / Disconnected from the profile's Instagram session cookie.

Supporting engine change: `MultiWorkerScraper` gained `pause()` / `resume()`, and `stop()` now releases paused workers.

Testing evidence

All 58 pytest tests pass (`pytest` in about 2 s); live headless runs against Instagram confirmed status detection, blocking and pause/resume.

Check	Result
<code>tests/test_ban_engine.py</code> (Sprint 1)	Ban linking, including the 4 required cases — pass
<code>tests/test_core_helpers.py</code> (Sprint 1)	Blocker rules, EN/TR dialog parsing, challenge and not-found detection, chunking — pass
<code>tests/test_gui_bridge.py</code> (Sprint 2, 15 tests)	Event forwarding, Start/Pause/Stop state flags, crash handling, session cookie check, real-window GUI smoke test — pass
Live run, logged out	<code>instagram</code> , <code>nasa</code> = ACTIVE; non-existent user = NOT_FOUND; Threads login wall = BLOCKED and run auto-stopped
Live run via GUI bridge	Paused 3 s with zero accounts processed, then resumed and finished all 3 accounts

Bugs found by testing and fixed before push:

- The prototype treated any URL containing "login" (e.g. user `loginking`) as a login wall; now matched by whole path segment.
- Instagram's current wording "Profile isn't available" was missing, so missing accounts showed as ACTIVE; now NOT_FOUND.
- A `stop()` issued just before a run started was silently cleared; now it always holds.

Deviations from spec for review

Five design decisions differ from the directives; each is deliberate and easy to reverse if preferred.

#	Spec said	Implemented	Reason
1	<code>checkpoint</code> → BANNED	A challenge on <i>our checker session</i> → BLOCKED, and the run stops	Otherwise a challenge on our own account would mark every target account as banned.
2	5 isolated browser contexts	1 persistent context with 5 pages	Only one persistent context can use the logged-in profile; all workers must share the session.
3	Colour badge on the Status cell	Whole row tinted by status	CustomTkinter has no table widget; <code>ttk.Treeview</code> can only colour whole rows.
4	Setup button runs <code>python -m src.core.scrapers --login</code>	Calls the login function on a background thread	<code>python -m</code> will not exist inside the packaged <code>.exe</code> (Sprint 5).
5	Header shows "Connected: @username"	Shows Connected / Disconnected only	Reading the username needs the encrypted cookie value; we only read cookie names and expiry.

Known gaps and risks

The largest gap is that the client's core data (country, joined date) has never been extracted live, because no checker account was available.

- **Logged-in extraction untested.** Meta only shows "About this account" to logged-in users. The dialog parsing is unit-tested on sample text, but the menu clicks and selectors are unverified on real pages.
- **Banned vs not found on Instagram.** Instagram web shows the same "Profile isn't available" page for banned and deleted accounts, so both currently report NOT_FOUND. A ban can only be confirmed when Threads reports it.
- **Speed target unverified.** 100 accounts in 4–5 minutes (TSK-401) has not been benchmarked.
- **Rate limits and Meta's terms.** Automated checking can trigger challenges on the checker account. The engine stops on a challenge, but a disposable account is essential.
- **Threads menu detection** relies on SVG coordinates at a 1280 px viewport and may break if Meta changes the layout.

Next steps

Needed first from Huzaifah: review and merge the Sprint 2 PR, and provide a disposable checker account for live testing.

- ☐ Review and merge Sprint 2 PR into `dev`
- ☐ Provide a disposable Instagram/Threads checker account
- ☐ Live-test country and joined-date extraction on 10 IG and 10 Threads accounts (TSK-102/103 acceptance)
- ☐ Sprint 3: Google Sheets TSV copy, toast feedback, SQLite history, CSV/Excel export (TSK-301 to 305)
- ☐ Sprint 4: 100-account benchmark, edge cases, checkpoint handling (TSK-401 to 404)
- ☐ Sprint 5: PyInstaller `.exe`, clean-machine test, user guide, handover (TSK-501 to 504)